UX PORTFOLIO

MALLORY GRIMES



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Hi!

As a **pioneer of User Experience (UX)**, I am proficient at building a UX culture and excel at applying Design Thinking and User Centered Design methodologies to atypical projects.

Furthermore, my **experience as a software developer** enables me to create technically feasible designs and ensures such designs are implemented as advertised.

This portfolio demonstrates my ability to enter diverse, deeply complex environments and make an impact.

- Mallory

EXPERTISE

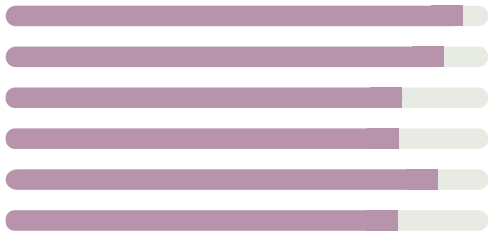
DESIGN



DEVELOPMENT



PERSONAL



case studies



C5ISRT
Autonomous Dashboard



SIGINT
Audio/Visual Feature



3D Engineering
Mobile Application

C5ISRT AUTONOMOUS DASHBOARD



CHALLENGES

How might we create an application that allows users to monitor and manage autonomous tasking and control of systems?

How might we bridge the gap between the processes and systems in place today and the massive paradigm shift the Department of War (DOW) is currently planning for future warfare?

SOLUTION

Delivered a visual dashboard that allows users, who are under heavy cognitive loads and in life threatening situations, to quickly observe the state of their systems/environment and provided intuitive means to take authoritative control as needed.

IMPACT

Demonstrated, in front of 70+ distinguished visitors including 4-star generals, that these novel, autonomous processes could successfully be integrated with existing systems, which secured stakeholder buy-in across multiple military branches and propelled system participation by over 150%.

ROLE

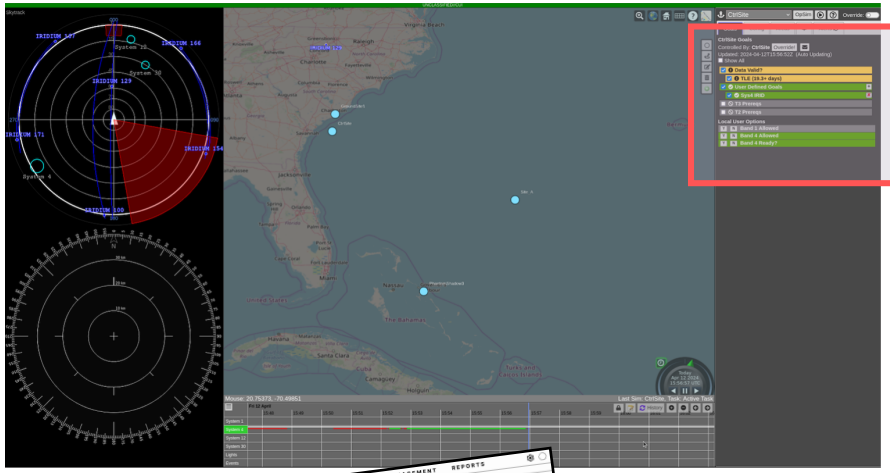
Managing the lack of contact and complete ambiguity around future user roles, I instead persistently pursued a depth of knowledge in both present processes and future expectations. Regardless of who would be involved, my understanding of the project domain including where touchpoints were, the fault tolerance of automation, and intervention requirements directly shaped the interface and influenced product strategy at the greater DOW level.

I empathized with the concerns of our forces regarding autonomy and authoritative control and created an intuitive data architecture that met the user's need for cognitive clarity and stakeholder's demands for maximum data visibility.

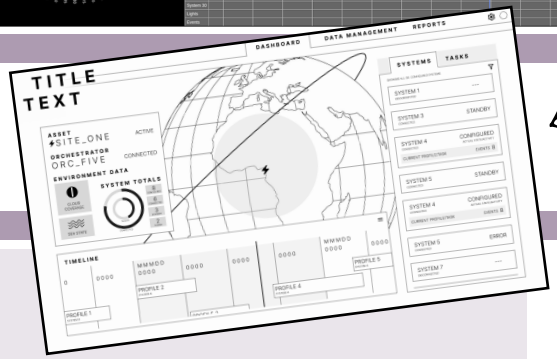
As the lone UX resource, I owned the end-to-end design lifecycle, collaborating occasionally with an external specialized group.

Bridging the gap between strategy and execution, I also contributed as Lead UI Developer ensuring mockups were implemented as advertised by working on features and themes where needed and guiding junior developers.

LEGACY SOFTWARE



- Core functionality is less than 1/3 of the screen.
- Very textual/data laden - little conveyed at a glance
- Inconsistent use of status colors
- Required extensive training to effectively use



Wireframe

DELIVERED MOCKUP/SOLUTION



- Highly visual; achieving cognitive clarity
- Branded; establishing confidence and trust in the product
- Extensive data architecture including over 17 consistent state varieties across task and systems

SIGINT AUDIOVISUAL FEATURE



CHALLENGES

How might we combat the lack of storage on a software defined radio (SDR) that of which limits tactical collections like how long a recording can be and/or what bandwidth can be captured for signals intelligence (SIGINT)?

How might we enable signal collectors to evaluate the validity and relevance of recordings in the field for signal analysts?

How might we enable collectors to reduce data files to meet the size requirements of their limited communication networks without compromising qualitative and quantitative data for analysts?

SOLUTION

A SIGINT recording feature that leveraged audiovisual radio frequency (RF) spectrums and tools and image editing capabilities to maintain a unified data structure with different contextual “views” to support multiple users and needs.

IMPACT

Transformed a previous, time-consuming auditory task into a sub-second, visual evaluation process allowing collectors to quickly evaluate recordings and remove obsolete data. Furthermore, cropping recordings around areas of interest instantly reduced file sizes. The reduced data volume maximized tactical collection capabilities for signal collectors while out on operations.

ROLE

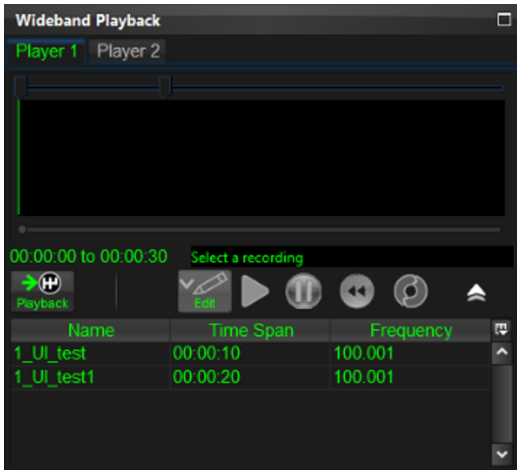
Solely accountable for user experience and user interface design, I traced the holistic lifecycle of SIGINT records collection and interviewed a variety of users involved end-to-end gaining valuable environmental context.

I audited other SDRs, including mobile handheld solutions used for quick short missions, to establish an operational baseline and identify critical gaps and bottlenecks.

My extensive research, wireframes, and user testing, coupled with collaboration from developers and RF engineers resulted in analogous innovation.

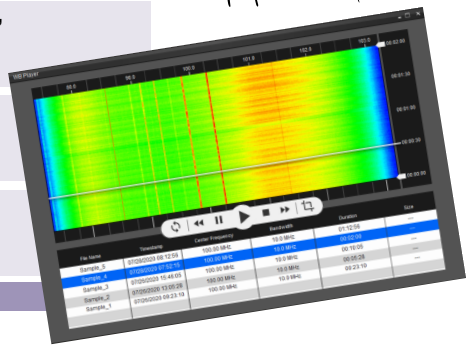
Orchestrated the technical implementation of this proven solution and supported training and fielding efforts.

EXISTING CAPABILITY



- ZERO understanding of content within listed recordings
- No concept of current file sizes/storage
- Ambiguous redundancy between “Playback” and “Play” button/functionality
- Inability to trim white noise or obsolete data
- Requires listening and scribing times to document relevance

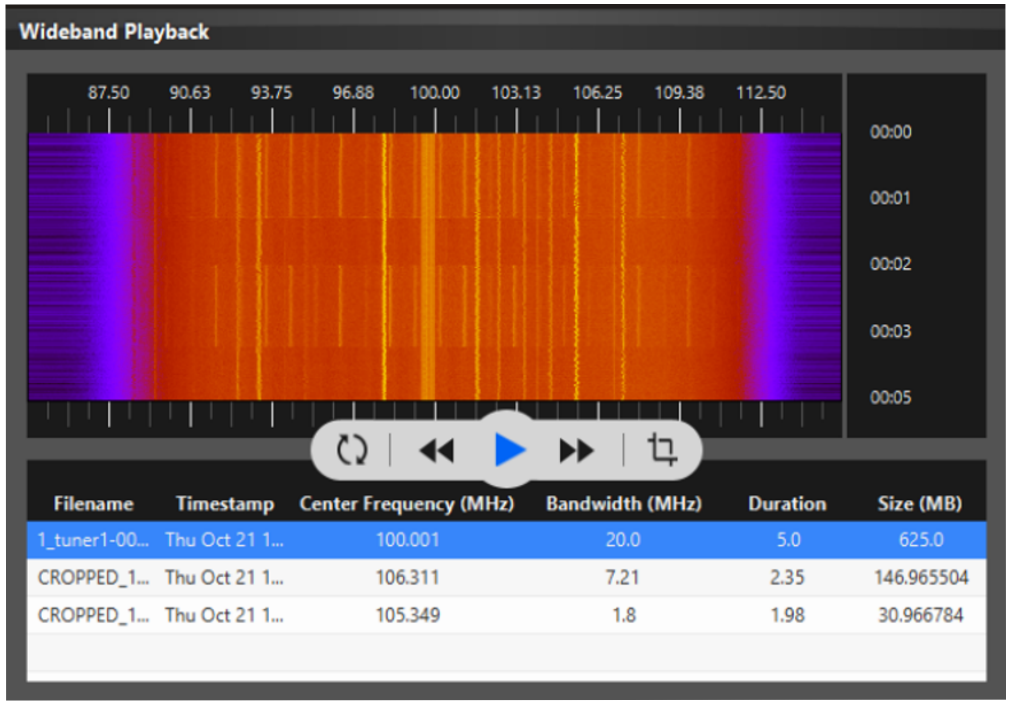
MOCKUP



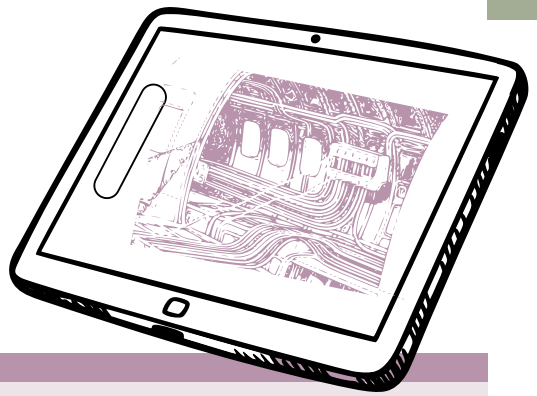
FIELDED

IMPLEMENTED SOLUTION

- Immediate validation of relevance upon hover
- Clear affordance regarding audio media and playback
- More details regarding data volume and storage needs
- Image cropping capabilities to quickly capture relevant data only



3D ENGINEERING MOBILE APPLICATION



CHALLENGES

Driven by the company's digital transformation, how might we utilize 3D engineering models for wiring installation?

How might these complex, 3D models be consumed on the manufacturing line?

How might we create a solution that encompasses many different types of 2D wiring drawings and supports a variety of user roles across a highly distributed and culturally different production ecosystem?

SOLUTION

A 3D portable document format (PDF), tailored to mobile tablets, which contained the model itself and provided useful tools that directly supported wiring installation/testing for a variety of different wiring parts across production sites.

IMPACT

Championing incremental innovation, these 3D PDFs preserved legacy infrastructure while unlocking next-generation technical capabilities. Incorporating optimized and enhanced workflows drastically accelerated time to value (TTV).

NO IMAGES AVAILABLE

ROLE

Serving as the only UX Specialist on a SAFe Agile team, I extensively evaluated 15+ different types of 2D wiring drawings. I interviewed design engineers, wiring technicians, and auditors across multiple production sites and manufacturing lines where culture varied greatly.

These deep dives produced an invaluable, quantitative breadth of knowledge. However, my time and experience in the plane on the manufacturing line provided the most profound qualitative empathy.

I synthesized this complex research for our team and leveraged it to lead cross-functional design studios, deliver mockups, and execute rigorous user testing cycles.

Leading design throughout the product lifecycle, I partnered with the Product Owner on product roadmaps, supported the scrum master as validated designs were broken down into stories, and collaborated with the engineering team to unblock and support developers during implementation.



TAKEAWAYS

I have **proven ability to rapidly master complex, unfamiliar industries** and deliver immediate UX value

Im **adept at identifying optimization opportunities** across digital products, operational workflows, and other complex organizational challenges

I leverage my **strong developer background** to design feasible solutions, facilitate cross-functional communication, and directly contribute to the codebase

I am a **full-spectrum creative artist** capable of not only desiging mockups, but establishing brands and delivering a variety of graphic content

And I **empathetically advocate for UX**, championing user needs and motivating my team to deliver exceptional, high quality solutions

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